

Load-driven relative-pitch-rate model

FBD → measured suspension force → normalized moment balance → rate predictor

1 Measure suspension loading

$$m_p a_p = m_p g + R + T + D_p$$

$$f_p = a_p - g, \quad \tau = T/m_p, \quad d_p = D_p/m_p$$

$$R/m_p = f_p - \tau e_x - d_p$$

Gravity cancels. No canopy lift/drag model is needed here.

2 Add pitch moments: internal hinge torque cancels

$$I_p \dot{q}_p = (-a \times R)_y + z_T T + J$$

$$I_c \dot{q}_c = (b \times R)_y + M_a - J$$

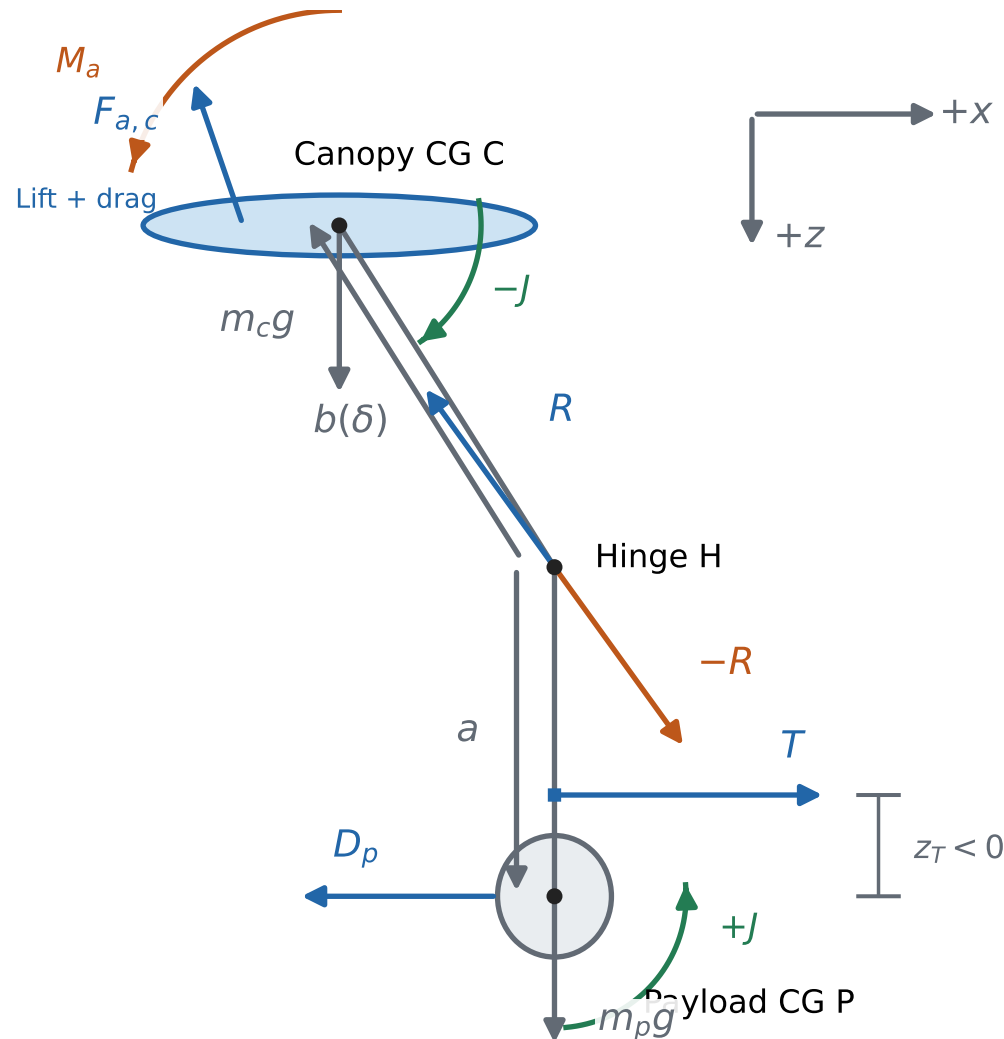
$$\kappa_p = I_p/m_p, \quad \kappa_c = I_c/m_p, \quad m_a = M_a/m_p$$

$$\begin{aligned} \kappa_c \dot{q}_c + \kappa_p \dot{q}_p \\ = [(b - a) \times (f_p - \tau e_x - d_p)]_y + z_T \tau + m_a \end{aligned}$$

3 Integrate a momentum state instead of differentiating gyro

$$s = \kappa_c q_c + \kappa_p q_p, \quad \dot{s} = \text{RHS above}$$

$$q_{\text{rel}} = (s - \kappa_p q_p)/\kappa_c - q_p$$



J = hinge torque on payload; $-J$ acts on canopy. M_a = external canopy aerodynamic pitch moment and does not cancel. q_p and q_c are payload and canopy pitch rates; $q_{\text{rel}} = q_c - q_p$. All vectors use a common payload frame. Force arrows are schematic. This is a planar propagation model; an observer still needs measurement corrections to reject drift and model error.